Computer Networks — Lecture 01: Application Layer (Principles of Network Applications)

ية الشبكة

Part 1: Multiple Choice (25)

1. Which statement correctly distinguishes application architecture from network architecture?

أي جملة بتفرّق صح بين معمارية التطبيق ومعمارية الشبكة؟

A. Both are designed by the application developer for each new application
B. The network architecture is designed by the developer, while the application architecture is fixed
C. The application architecture is designed by the developer, while the network architecture is fixed from the developer's perspective
D. Both are fixed and the developer can only choose between them

جابة الصحيحة:

C. The application architecture is designed by the developer, while the network architecture is fixed from the developer's perspective

◆ ليه صح:

يصمّمها المطوّر وبتحدد إزاي التطبيق متوزع على الأجهزة، أما معمارية الشبكة (الخمس طبقات)
وبتقدم خدمات للتطبيقات. باقي الاختيارات بتعكس الأدوار أو بتعتبر الاتنين ثابتين أو الاتنين من تصميم المطوّر
وده غلط.

▶ Ref: Lecture 01 — Section 2.1.1

2. On which devices does a developer write and run application software?

المطوّر بيكتب ويشغّل برامج التطبيق على أنهي أجهزة؟

A. Only on end systems
B. On routers and link-layer switches
C. On all network-core devices
D. On both end systems and routers

جابة الصحيحة:

A. Only on end systems

◆ ليه صح:

ي الـ routers والـ switches. التصميم ده سهّل تطوير التطبيقات بسرعة.

▶ Ref: Lecture 01 — Section 2.1

3. Which statement about clients in a client-server architecture is TRUE?

ن الـ clients في معمارية client-server؟

A. Clients must be always-on with a fixed IP address
B. Clients communicate directly with each other to share the load
C. Clients are housed in data centers
D. Clients do not communicate directly with each other

جابة الصحيحة:

D. Clients do not communicate directly with each other

◆ ليه صح:

always-on والعنوان الثابت من خصائص السيرفر مش الـ client، والاتصال المباشر بين الأجهزة من خصائص P2P.

▶ Ref: Lecture 01 — Section 2.1.1

4. Which pair of properties describes the server in a client-server architecture?

في client-server؟

A. Intermittently connected host with a changing address
B. Always-on host with a fixed, well-known IP address
C. Host owned and controlled by home users
D. Host that never receives requests from clients

جابة الصحيحة:

B. Always-on host with a fixed, well-known IP address

◆ ليه صح:

ي الـ peers في P2P.

▶ Ref: Lecture 01 — Section 2.1.1

5. Why is a data center often used in a client-server application?

خدم data center في تطبيقات client-server؟

A. Because clients need to talk directly to each other
B. A single server host cannot keep up with all client requests, so many hosts form a powerful virtual server
C. Because data centers remove the need for IP addresses
D. Because P2P applications require them

جابة الصحيحة:

B. A single server host cannot keep up with all client requests, so many hosts form a powerful virtual server

◆ ليه صح:

لأجهزة data center
راضي قوي. باقي الاختيارات مالهاش علاقة بسبب استخدامه.

▶ Ref: Lecture 01 — Section 2.1.1

6. Which application is given as a popular example of a P2P architecture?

ارية P2P؟

A. The Web
B. FTP
C. Telnet
D. BitTorrent

جابة الصحيحة:

D. BitTorrent

◆ ليه صح:

BitTorrent هو المثال المذكور لـ file sharing وFTP وTelnet كلها أمثلة لـ client-server.

▶ Ref: Lecture 01 — Section 2.1.1

7. What does the self-scalability of P2P architectures mean?

ى الـ self-scalability في معمارية P2P؟

A. Each peer adds service capacity to the system while also generating workload
B. Peers never request files from other peers
C. The system relies on one powerful dedicated server
D. Server bandwidth grows automatically in data centers

جابة الصحيحة:

A. Each peer adds service capacity to the system while also generating workload

◆ ليه صح:

كس P2P.

▶ Ref: Lecture 01 — Section 2.1.1

8. Which challenges do P2P applications face because of their highly decentralized structure?

؟P2P

A. High server infrastructure costs and server bandwidth
B. The need for an always-on server with a fixed address
C. Security, performance, and reliability
D. Inability of peers to communicate directly

جابة الصحيحة:

C. Security, performance, and reliability

◆ ليه صح:

والـ bandwidth ميزة P2P بتفاداها أصلًا، وكمان الـ peers
صل ببعض مباشرة.

▶ Ref: Lecture 01 — Section 2.1.1

9. Which description fits the peers in a P2P architecture?

ف الـ peers في معمارية P2P؟

A. Always-on servers owned by the service provider
B. Routers located in the network core
C. Dedicated hosts in a data center
D. Intermittently connected desktops and laptops controlled by users in homes, universities, and offices

جابة الصحيحة:

D. Intermittently connected desktops and laptops controlled by users in homes, universities, and offices

◆ ليه صح:

والـ data centers خاصة بـ client-server.

▶ Ref: Lecture 01 — Section 2.1.1

10. In operating-system jargon, what actually communicates across a network?

مصطلحات نظم التشغيل، إيه اللي بيتواصل فعليًا عبر الشبكة؟

A. Programs that are not running
B. Sockets
C. Processes
D. Protocol layers

جابة الصحيحة:

C. Processes

◆ ليه صح:

و الـ processes (البرنامج وهو شغال جوه الـ end system) مش البرامج نفسها. الـ socket
يا الـ process.

▶ Ref: Lecture 01 — Section 2.1.2

11. How do processes running on two different end systems communicate?

يظ

A. By exchanging messages across the computer network
B. By interprocess communication governed by the operating system
C. By sharing the same memory space
D. By directly connecting their sockets without any network

جابة الصحيحة:

A. By exchanging messages across the computer network

◆ ليه صح:

ائل (messages) عبر الشبكة. الـ
بالـ processes على نفس الجهاز، ومش محل اهتمام المحاضرة.

▶ Ref: Lecture 01 — Section 2.1.2

12. In a communication session, which process is labeled the client?

، أي process بيتسمى client؟

A. The process that waits to be contacted
B. The process that initiates the communication
C. The process running on the faster host
D. The process that uploads the file

جابة الصحيحة:

B. The process that initiates the communication

◆ ليه صح:

الـ client هو اللي بيبدأ الاتصال ويتواصل أول مرة مع
uploading peer في P2P هو الـ server مش الـ client.

▶ Ref: Lecture 01 — Section 2.1.2

13. In P2P file sharing, Peer A asks Peer B to send a specific file. What are their roles in this session?

في P2P لو Peer A

A. Peer A is the server and Peer B is the client
B. Both peers are servers
C. Peer A is the client and Peer B is the server
D. Neither is a client or a server

جابة الصحيحة:

C. Peer A is the client and Peer B is the server

◆ ليه صح:

A هو اللي بدأ الطلب فهو الـ client، وB اللي بيبعت الملف فهو الـ server في الجلسة دي. نفس الـ peer
في جلسة server و

▶ Ref: Lecture 01 — Section 2.1.2

14. What is a socket?

و الـ socket؟

A. The software interface between the application layer and the transport layer within a host
B. The physical link between two hosts
C. The interface between the network layer and the link layer
D. A router port that forwards packets

جابة الصحيحة:

A. The software interface between the application layer and the transport layer within a host

◆ ليه صح:

الـ socket هو واجهة برمجية بين طبقة التطبيق وطبقة النقل جوه نفس الجهاز، ويتسمى كمان
والـ link.

▶ Ref: Lecture 01 — Section 2.1.2

15. In the house-and-door analogy, what does the socket correspond to?

، البيت والباب؟ socket

A. The house itself
B. The postal address
C. The road between houses
D. The door of the house

جابة الصحيحة:

D. The door of the house

◆ ليه صح:

ه. الـ socket هو البيت والباب
الأبواب.

▶ Ref: Lecture 01 — Section 2.1.2

16. What control does the developer have on the transport-layer side of the socket?

ة الـ transport layer من الـ socket؟

A. Complete control over how the transport protocol is implemented
B. Choosing the transport protocol and perhaps fixing a few parameters such as maximum buffer and segment sizes
C. Choosing the route packets take through the network
D. No control at all, even over protocol choice

جابة الصحيحة:

B. Choosing the transport protocol and perhaps fixing a few parameters such as maximum buffer and segment sizes

◆ ليه صح:

موية parameters زي أقصى buffer و segment size
كم عند نظام التشغيل.

▶ Ref: Lecture 01 — Section 2.1.2

17. Which two pieces of information identify the receiving process on the Internet?

إلى process؟

A. The IP address of the host and a destination port number
B. The host name and the socket buffer size
C. The IP address and the transport protocol name
D. The port number and the application's file name

جابة الصحيحة:

A. The IP address of the host and a destination port number

◆ ليه صح:

ن الـ IP بيحدد الجهاز ورقم الـ port بيحدد البرنامج اللي جواه.
تطبيقات كتير.

▶ Ref: Lecture 01 — Section 2.1.2

18. According to the lecture, an IP address is:

ن الـ IP هو:

A. A 16-bit quantity that identifies a process
B. A 48-bit quantity that identifies a socket
C. A 128-bit quantity that identifies a web page
D. A 32-bit quantity that uniquely identifies a host

جابة الصحيحة:

D. A 32-bit quantity that uniquely identifies a host

◆ ليه صح:

يد الـ IP address كويس جدًا 32 bit
مش process.

▶ Ref: Lecture 01 — Section 2.1.2

19. Which pair of well-known port numbers is correct according to the lecture?

م المحاضرة؟

A. Web server: 25, SMTP mail server: 80
B. Web server: 80, SMTP mail server: 25
C. Web server: 80, SMTP mail server: 80
D. Web server: 25, SMTP mail server: 25

جابة الصحيحة:

B. Web server: 80, SMTP mail server: 25

◆ ليه صح:

خدم port 80 وسيرفر الإيميل (SMTP) بيستخدم 25 أو
رها.

▶ Ref: Lecture 01 — Section 2.1.2

20. Which set lists the four dimensions of transport-layer services?

ت الـ transport layer؟

A. Reliability, routing, addressing, and timing
B. Throughput, encryption, switching, and security
C. Reliable data transfer, throughput, timing, and security
D. Reliable data transfer, routing, timing, and framing

جابة الصحيحة:

C. Reliable data transfer, throughput, timing, and security

◆ ليه صح:

هي: reliable data transfer وthroughput وtiming وsecurity. الـ routing والـ addressing والـ framing مش من خدمات النقل المذكورة هنا.

▶ Ref: Lecture 01 — Section 2.1.3

21. What does a reliable data transfer service guarantee?

دمة reliable data transfer؟

A. Data always arrives within 100 msec
B. Data sent by one process is delivered correctly and completely to the other process
C. The sender always gets at least a fixed throughput
D. All transmitted data is encrypted

جابة الصحيحة:

B. Data sent by one process is delivered correctly and completely to the other process

◆ ليه صح:

ة للـ process التاني. ضمان زمن 100 msec وضمان معدل throughput، والتشفير ده security.

▶ Ref: Lecture 01 — Section 2.1.3

22. Which kind of applications can tolerate some data loss?

أنهي نوع من التطبيقات ممكن يتحمل فقدان جزء من البيانات؟

A. Electronic mail
B. Financial applications
C. Loss-tolerant multimedia applications such as conversational audio/video
D. File transfer

جابة الصحيحة:

C. Loss-tolerant multimedia applications such as conversational audio/video

◆ ليه صح:

لملفات والبيانات
لمالية محتاجة نقل موثوق لأن فقد البيانات ليه عواقب سيئة.

▶ Ref: Lecture 01 — Section 2.1.3

23. Which of the following is an elastic application?

عتبر elastic application؟

A. Internet telephony that needs 32 kbps
B. A multimedia application with a fixed required throughput
C. Any bandwidth-sensitive application
D. Web transfers

جابة الصحيحة:

D. Web transfers

◆ ليه صح:

الـ elastic applications (زي الإيميل ونقل الملفات والويب) بتستفيد بأي قدر من الـ throughput المتاح، أو
لب throughput محدد اسمها bandwidth-sensitive.

▶ Ref: Lecture 01 — Section 2.1.3

24. Which type of application benefits most from a guarantee that data arrives no more than 100 msec later?

من 100 msec؟

A. Interactive real-time applications such as multiplayer games
B. Electronic mail
C. File transfer
D. Web document transfer

جابة الصحيحة:

A. Interactive real-time applications such as multiplayer games

◆ ليه صح:

علية real-time (الاتصال الصوتي، الألعاب الجماعية، teleconferencing) محتاجة قيود زمن صارمة.
ر الـ real-time كل اللي بتفضّله إن التأخير يبقى أقل، ومفيش قيد صارم.

▶ Ref: Lecture 01 — Section 2.1.3

25. In the sending host, a transport protocol encrypts data and the receiving host decrypts it. Which security service does this provide?

وده بيوفر أمن إيه؟ transport protocol

A. Guaranteed throughput
B. Timing guarantee
C. Confidentiality
D. Reliable data transfer

جابة الصحيحة:

C. Confidentiality

◆ ليه صح:
ن الـ confidentiality، يعني حتى لو حد راقب البيانات في الطريق مش هيفهمها. الـ transport
قدم data integrity و end-point authentication.

▶ Ref: Lecture 01 — Section 2.1.3

Part 2: Essay Questions (10)

1. Explain the difference between application architecture and network architecture, and name the two predominant application architectures.

بين معمارية التطبيق ومعمارية الشبكة، واذكر المعماريتين الأساسيتين للتطبيقات.

Model answer:
• The network architecture (e.g., the five-layer Internet architecture) is fixed and provides a specific set of services to applications.
• The application architecture is designed by the application developer and dictates how the application is structured over the various end systems.
• The two predominant paradigms are the client-server architecture and the peer-to-peer (P2P) architecture.
▶ Ref: Lecture 01 — Section 2.1.1

2. Describe the client-server architecture and its main characteristics. Give examples of applications that use it.

ارية client-server وخصائصها الأساسية، مع أمثلة على تطبيقات بتستخدمها.

Model answer:
• There is an always-on host called the server that services requests from many other hosts called clients.
• Clients do not communicate directly with each other (e.g., two browsers do not talk directly).
• The server has a fixed, well-known IP address, so a client can always contact it by sending a packet to that address.
• Examples: the Web, FTP, Telnet, and e-mail.
▶ Ref: Lecture 01 — Section 2.1.1

3. Why are data centers used in client-server applications? Give examples of services that run in data centers.

م الـ data centers في تطبيقات client-server؟ اذكر أمثلة لخدمات بتشتغل فيها.

Model answer:
• A single server host is often incapable of keeping up with all client requests; for example, a popular social-networking site would be overwhelmed by one server.
• A data center houses a large number of hosts that together create a powerful virtual server.
• Examples: search engines (Google, Bing, Baidu), Internet commerce (Amazon, eBay, Alibaba), Web-based e-mail (Gmail, Yahoo Mail), and social media (Facebook, Instagram, Twitter, WeChat).
▶ Ref: Lecture 01 — Section 2.1.1

4. Describe the P2P architecture. What are its advantages and challenges?

ا وتحدياتها؟ P2P

Model answer:
• P2P has minimal (or no) reliance on dedicated servers; the application exploits direct communication between pairs of intermittently connected hosts called peers.
• Peers are desktops and laptops controlled by users, mostly in homes, universities, and offices; an example is BitTorrent.
• Advantage 1: self-scalability, because each peer generates workload by requesting files but also adds service capacity by distributing files.
• Advantage 2: cost effectiveness, since significant server infrastructure and server bandwidth are not required.
• Challenges: security, performance, and reliability, due to the highly decentralized structure.
▶ Ref: Lecture 01 — Section 2.1.1

5. Compare the client-server and P2P architectures in terms of communication, server dependence, and scalability.

قارن بين معماريتي client-server وP2P من حيث الاتصال والاعتماد على السيرفر والقدرة على التوسع.

Model answer:

• Communication: in client-server, clients talk only to the server; in P2P, peers communicate directly with each other.

• Server dependence: client-server relies on an always-on server (often in a data center) with a fixed IP address; P2P has minimal or no reliance on dedicated servers.

• Scalability: client-server needs more servers or a data center to handle load; P2P is self-scaling because every peer adds capacity.

• Cost and issues: P2P is cost effective but faces security, performance, and reliability challenges.

▶ Ref: Lecture 01 — Section 2.1.1

6. Define client and server processes. Can one process be both? Explain with examples from the Web and P2P file sharing.

عرّف الـ client والـ server processes. هل يمكن أن يكون process واحد هو الاثنين؟ اشرح بأمثلة من الويب وP2P.

Model answer:

• In the context of a communication session, the process that initiates the communication is the client; the process that waits to be contacted to begin the session is the server.

• Web: the browser process initiates contact with the Web server process, so the browser is the client and the Web server is the server.

• P2P: when Peer A asks Peer B for a file, Peer A is the client and Peer B is the server for that session.

• A process in P2P can be both a client and a server (it can both upload and download), but within any given session one is labeled the client and the other the server.

▶ Ref: Lecture 01 — Section 2.1.2

7. What is a socket? Explain the house-and-door analogy and the developer's control on each side of the socket.

ما هي الـ socket؟ اشرح تشبيه البيت والباب، وما الذي يتحكم به المطوّر على كل جانب من الـ socket.

Model answer:

• A socket is the software interface through which a process sends messages into and receives messages from the network; it lies between the application layer and the transport layer within a host and is also called the API between the application and the network.

• Analogy: the process is like a house and the socket is its door. The sender pushes the message out of its door, assuming a transportation infrastructure on the other side delivers it to the door of the destination process.

• The developer controls everything on the application-layer side of the socket.

• On the transport-layer side, the developer can only choose the transport protocol and perhaps fix a few parameters such as maximum buffer and maximum segment sizes.

▶ Ref: Lecture 01 — Section 2.1.2

8. Explain how a process is addressed in the Internet. Give examples of well-known port numbers.

كيف يتم عنونة الـ process على الإنترنت، مع أمثلة على أرقام ports معروفة.

Model answer:

• Two pieces of information are needed: the address of the host and an identifier that specifies the receiving process in the destination host.

• The host is identified by its IP address, a 32-bit quantity that uniquely identifies the host.

• A destination port number identifies the receiving process (more specifically, the receiving socket), because a host may run many network applications.

• Examples: a Web server is identified by port 80, and a mail server process using SMTP by port 25.

▶ Ref: Lecture 01 — Section 2.1.2

9. Explain the reliable data transfer and throughput services of a transport protocol. What are loss-tolerant, bandwidth-sensitive, and elastic applications?

اشرح خدمتي reliable data transfer وthroughput في الـ transport protocol. وإيه معنى loss-tolerant وbandwidth-sensitive وelastic؟

Model answer:

• Reliable data transfer: guarantees data sent by one end is delivered correctly and completely to the other end, so the sender can pass data into the socket with complete confidence. It is needed by e-mail, file transfer, remote host access, Web transfers, and financial applications.

• Loss-tolerant applications (e.g., conversational audio/video) can tolerate some data loss.

• Throughput: available throughput is the rate at which the sending process can deliver bits to the receiving process, and it can fluctuate because other sessions share the path. A protocol could guarantee a throughput of at least r bits/sec.

• Bandwidth-sensitive applications have specific throughput requirements (e.g., Internet telephony encoding voice at 32 kbps); some use adaptive coding to match available throughput.

• Elastic applications (e-mail, file transfer, Web transfers) can use as much or as little throughput as is available.

▶ Ref: Lecture 01 — Section 2.1.3

10. Explain the timing and security services that a transport-layer protocol can provide to applications.

ي الـ timing والـ security اللي ممكن الـ transport-layer protocol يقدمهم للتطبيقات.

Model answer:

• Timing: a protocol can guarantee, for example, that every bit sent into the socket arrives at the receiver's socket no more than 100 msec later.

• This suits interactive real-time applications such as Internet telephony, virtual environments, teleconferencing, and multiplayer games; long delays cause unnatural pauses or make the application feel less realistic.

• For non-real-time applications, lower delay is preferable but no tight constraint is placed on end-to-end delay.

• Security: the protocol can encrypt data at the sending host and decrypt it at the receiving host, providing confidentiality even if the data is observed in transit.

• It can also provide data integrity and end-point authentication (covered in Chapter 8).

▶ Ref: Lecture 01 — Section 2.1.3